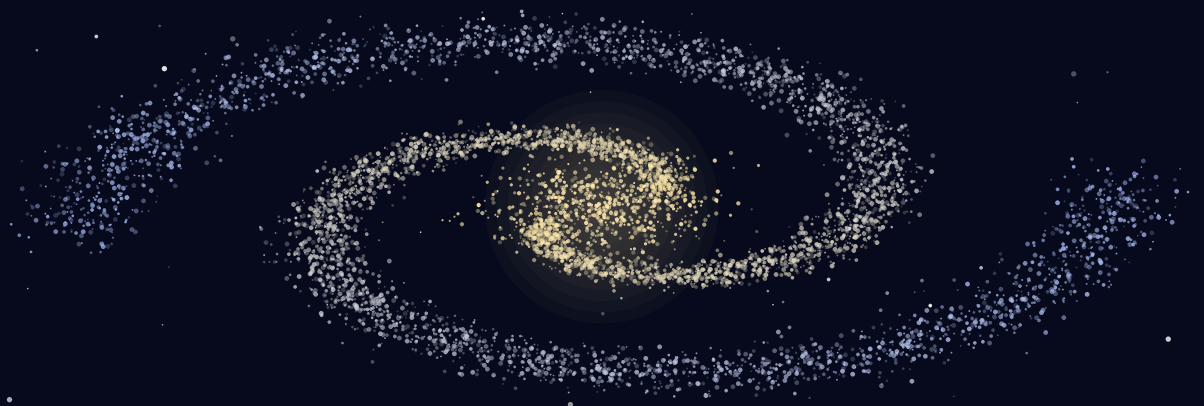


AN INTERACTIVE STORYTELLING JOURNEY

The Galactic Canvas

Milky Way / Solar System / Sun / Planets

Research Compilation - Expanded Edition

Merged research, target audience and two-page planet guides



Compiled September 2026 | Planet and moon counts updated to mid-2026

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PROJECT OVERVIEW

The Galactic Canvas is an interactive storytelling website that takes the viewer on a visual and narrative journey - beginning with the Milky Way galaxy, zooming into our Solar System, moving toward the Sun, and finally exploring each planet in detail. The site is a scroll- or zoom-based experience that guides users from the largest cosmic scale down to individual planetary facts.

WHAT IS NEW IN THIS EDITION

- **Stage 4 expanded:** every planet now has a two-page guide - a **Summary** page (quick facts, story, nine quick answers) and a **Learn More** page (interior, moon details, exploration timeline, fun facts).
- **Target audience added:** a detailed audience profile with needs, design implications and accessibility notes (Section 1).
- **Research merged:** the original Q&A answers and the quantitative/qualitative data tables now sit together under each stage, so nothing is repeated.
- **Moon counts refreshed to mid-2026:** Jupiter 101+ (was 95), Saturn 285+ (was 146), Uranus 29 (was 27), Neptune 16 (was 14); over 430 moons now orbit the eight planets. Exact totals differ slightly by source, so re-check before submission.
- **Small fixes:** Mercury's BepiColombo arrival date, Akatsuki's end of operations, planet diameter ratios, and a range for the galaxy count of the observable universe.

01 Purpose, Audience and Rationale

Why the site exists and who it is built for

WHY

Why are we using this website?

This website makes astronomy engaging and easy to understand through visual storytelling instead of plain text or static images. Space is vast and hard to grasp - by using an interactive zoom-based journey (Milky Way, Solar System, Sun, Planets), the site turns abstract scale and distance into something viewers can experience step by step. It combines education with an immersive, cinematic feel, helping users retain information better than a traditional textbook format, while also demonstrating skills in web design, interactivity and data storytelling.

- Simplify a complex topic (the scale of the universe) through visual, interactive design.
- Create an engaging, memorable learning experience rather than passive reading.
- Show how storytelling techniques (zoom, scroll, narrative flow) can teach science.
- Showcase creative and technical skills: design, animation and content research combined.

WHY THIS TOPIC

Why Milky Way and space?

- Space is universally fascinating and has strong visual potential, making it ideal for interactive storytelling.
- It naturally supports a zoom-in narrative - galaxy, star, planets - giving the project a clear beginning, middle and end.
- Reliable data (distances, sizes, compositions) is well documented, so research and fact-checking are straightforward.
- Striking visuals, animations and comparisons (scale, distance, temperature) keep users engaged.
- Strong educational value: the project can genuinely teach people something, beyond being an assignment.
- Personal interest in space and astronomy made the research and design process more motivating and enjoyable.

TARGET AUDIENCE

Who is the website for?

In one line: curious learners aged roughly 12 to 25 - school and college students - who want space explained visually, plus the teachers, space enthusiasts and evaluators who use or judge the site. Everyone is assumed to start with little or no astronomy background.

Primary audience	Students (school and college level), aged about 12-25, studying or exploring astronomy and science.
Secondary audience	Teachers and educators; space enthusiasts and self-paced learners.
Also served	Evaluators/examiners of this assignment; general visitors with little prior knowledge.
Prior knowledge	None to intermediate - terms such as 'light-year' or 'gas giant' are explained in plain language.
Devices	Smartphones and laptops; phone-first layout because many students browse on mobile, with light pages for slower connections.
Reading level	Plain English (about Grade 8). Summary pages stay short; deeper detail lives on the Learn More pages.
Motivation	Curiosity, homework and projects, classroom teaching, assessment, and simple enjoyment of space.

AUDIENCE SEGMENTS – WHAT THEY NEED AND HOW THE SITE RESPONDS

Segment	What they need	How The Galactic Canvas helps
Students (school and college)	An engaging way to learn; quick answers for homework; memorable visuals; a clear path from big ideas to facts.	Zoom-and-scroll story; one-page Summary per planet; 'Did you know?' hooks; Learn More for projects.
Teachers and educators	A visual aid that shows scale and the Solar System's structure; accurate, classroom-ready facts.	Large-to-small narrative order; scale comparisons; consistent page template; printable summaries; dated data and sources.
Space enthusiasts and curious learners	Depth, fresh numbers and stories, self-paced exploration.	Learn More pages with moon tables and mission timelines; moon counts shown with dates.
Evaluators / examiners	Accurate content, well-organised research, and polished interactive design.	Verified data tables; clear stages; consistent design system; smooth scroll animation.
General visitors	A hook, no jargon, nothing intimidating - they should enjoy it within seconds.	Cinematic opening, short plain-language captions, obvious 'scroll to zoom' cue.

EXAMPLE VISITORS (ILLUSTRATIVE PERSONAS)

Visitor	Goal	Likely path through the site
Aarav, 16 **Class 11 student**	Preparing a school science project on the Solar System and needs reliable facts fast.	Scrolls the zoom journey, opens the Mars Summary page, then Learn More for the moon and mission details to cite.
Ms. Mehta **Science teacher**	Wants a visual way to show the scale of space and compare planets in class.	Uses the Milky Way to Sun zoom as a class opener, then the planet comparison tables and scale checks.
Kabir, 24 **Space enthusiast**	Follows space news and likes up-to-date numbers and mission stories.	Goes straight to Learn More pages: Saturn's moon tally, Europa Clipper and Dragonfly timelines.

DESIGN IMPLICATIONS OF THE AUDIENCE

- **Progressive disclosure:** the two-page planet format lets casual visitors stop after the Summary while keen learners open Learn More.
- **Accessibility:** readable contrast and text size, alt text for visuals, keyboard-friendly scrolling, captions for any audio, and a reduced-motion option for zoom animations.
- **Clarity over jargon:** define each term the first time it appears and use familiar comparisons (for example, 'Earth is a grape next to a basketball-sized Jupiter').
- **Trust:** show dates next to fast-changing numbers such as moon counts, and keep a sources list for evaluators and teachers.

02 Research: Stages 1 to 3

The zoom-in story, with questions, answers and data together

STAGE 1 | ESTABLISHING SHOT

The Milky Way

What is the Milky Way, and what type of galaxy is it?	The Milky Way is the galaxy that contains our Solar System. Structurally it is a **barred spiral galaxy**: a central bar-shaped core with spiral arms winding outward.
How big is it?	Roughly **100,000-120,000 light-years** in diameter, containing an estimated **100-400 billion stars**.
Where is it, and what is nearby?	It belongs to a small cluster of galaxies called the **Local Group**. Its nearest large neighbour is the **Andromeda Galaxy**, with smaller satellites such as the Large and Small Magellanic Clouds.
Who discovered or named it, and how did ancient cultures see it?	Cultures worldwide described it as a river, road or trail of light. **Galileo Galilei** confirmed in 1610, with a telescope, that the band of light was countless individual stars. The name comes from the Latin 'Via Lactea', translating the Greek 'milky circle'.
How did it form, and how old is it?	About **13.6 billion years ago**, through the gravitational collapse and merging of gas clouds and smaller proto-galactic structures soon after the Big Bang.
What are its main parts?	A central bulge/core; spiral arms holding most visible stars; a flat rotating disk; and a surrounding halo of older stars, gas and dark matter.
Why is it a band of light in our sky?	Our Solar System sits inside the flat disk. Looking along the disk at night, we see the combined light of countless distant stars packed together as a hazy band.
What is at the centre?	**Sagittarius A***, a supermassive black hole about **4 million times the mass of the Sun**, around which the galaxy rotates.

STAGE 1 - BY THE NUMBERS AND STORY NOTES

- **Diameter:** ~100,000-120,000 light-years
- **Stars:** 100-400 billion
- **Age:** ~13.6 billion years
- **Sun's distance from the centre:** ~26,000 light-years
- **Disk thickness:** ~1,000 light-years
- **Orbital speed:** ~828,000 km/h; one orbit = ~230 million years (a 'galactic year')
- **Galaxies in the observable universe:** estimates range from ~200 billion to ~2 trillion
- Classified as a **barred spiral galaxy**.
- Made of a central bulge, spiral arms, a halo and dark matter.
- Contains gas, dust and stellar nurseries where new stars form.
- Named for its milky band (Latin: Via Lactea).
- Part of the Local Group; moving toward Andromeda. Older models predicted a merger in billions of years, but a recent study suggests it is far from certain.

STAGE 2 | ZOOMING IN

The Solar System

What is the Solar System, and where is it?	The Sun plus everything gravitationally bound to it - planets, moons, asteroids and comets. It lies in the **Orion Arm** of the Milky Way, about **26,000 light-years** from the galactic centre.
How did it form?	By the **nebular hypothesis**: about **4.6 billion years ago** a giant molecular cloud of gas and dust collapsed under gravity into a spinning disk. The Sun formed at the dense centre; the rest clumped into planets, moons and other bodies.
How many objects are in it?	**8 planets**, **5 officially recognised dwarf planets**, **over 430 moons** orbiting the planets (mid-2026 count), an estimated one to two million asteroids larger than 1 km, and countless comets.

Why do planets orbit the Sun?	Gravity constantly pulls each planet inward while its forward motion carries it onward; the balance produces a stable, slightly elliptical orbit.
What holds it together?	The Sun's gravity, from its huge mass - **99.86%** of the Solar System's total.
How long does it take to orbit the galaxy?	About **230 million years** - a 'galactic year'.

STAGE 2 – BY THE NUMBERS AND STORY NOTES

- **Age:** ~4.6 billion years
- **Planets:** 8 (plus 5 recognised dwarf planets)
- **Extent (to the Oort Cloud):** ~2 light-years
- **Moons of the planets:** over 430 (mid-2026)
- **Asteroids larger than 1 km:** an estimated 1-2 million
- Formed from a collapsing giant molecular cloud.
- Structure: inner rocky planets, asteroid belt, outer gas and ice giants, Kuiper Belt, Oort Cloud.
- Held together by the Sun's gravity.
- Planets orbit in roughly the same plane (the ecliptic) and direction.

STAGE 3 | THE ZOOM TARGET

The Sun

What is it made of, and what type of star is it?	Roughly **73% hydrogen** and **25% helium** plus trace heavier elements. A **G-type main-sequence star**, nicknamed a 'yellow dwarf'.
How does it produce energy?	By **nuclear fusion** in its core, where immense pressure and heat fuse hydrogen into helium, releasing energy as light and heat.
How old is it, and what is its future?	About **4.6 billion years old**, roughly midway through a main-sequence life of about 10 billion years. It will expand into a red giant and then shrink into a white dwarf.
How hot is it?	Core about **15 million °C**; visible surface (photosphere) about **5,500°C**.
Why does it matter for life on Earth?	Its light and heat drive Earth's climate, weather and water cycle, and power photosynthesis - the base of nearly all food chains.
What are solar phenomena?	**Sunspots** are cooler, darker regions caused by magnetic activity; **solar flares** are sudden bursts of energy and radiation; **solar wind** is a continuous stream of charged particles flowing into space.

STAGE 3 – BY THE NUMBERS AND STORY NOTES

- **Diameter:** ~1.39 million km (109 x Earth's)
- **Mass:** 99.86% of the Solar System's total
- **Surface temperature:** ~5,500°C
- **Core temperature:** ~15 million °C
- **Age and lifespan:** ~4.6 billion years; ~10 billion total
- **Composition:** ~73% hydrogen, ~25% helium
- G-type main-sequence star ('yellow dwarf').
- Powered by nuclear fusion (hydrogen into helium).
- Shows sunspots, solar flares and solar wind.
- Will become a red giant, then a white dwarf.

03 Stage 4: The Planets

Eight planets, each with a Summary page and a Learn More page

Stage 4 is the destination of the zoom. Each planet gets **two pages** that map directly to the website: a short **Summary** view that appears as the viewer arrives, and a **Learn More** view that opens for people who want to dig deeper.

PAGE 1 | SUMMARY

- Planet at a glance: 8 key facts
- The story in brief
- Nine quick answers (composition, orbit, day, size, moons, atmosphere, uniqueness, discovery, missions)
- A 'Did you know?' hook

PAGE 2 | LEARN MORE

- Inside and out: interior and surface features
- Moon details (or why there are none)
- Exploration timeline: missions and discoveries
- Fun facts, a website interaction idea and a scale check

QUANTITATIVE REFERENCE TABLE

Planet	Type	Distance from Sun (M km)	Diameter (km)	Day length	Year length	Moons (2026)
Mercury	Rocky	57.9	4,879	59 days	88 days	0
Venus	Rocky	108.2	12,104	243 days (retro.)	225 days	0
Earth	Rocky	149.6	12,742	24 hrs	365.25 days	1
Mars	Rocky	227.9	6,779	24.6 hrs	687 days	2
Jupiter	Gas giant	778.5	139,820	9.9 hrs	11.9 years	101+
Saturn	Gas giant	1,434	116,460	10.7 hrs	29.5 years	285+
Uranus	Ice giant	2,871	50,724	17.2 hrs	84 years	29
Neptune	Ice giant	4,495	49,244	16.1 hrs	165 years	16

Moon counts change often. New moons are confirmed every year, mostly tiny irregular ones around the giant planets. Counts here follow 2026 announcements (Saturn 285 confirmed in March 2026; Jupiter 101) - some trackers already list more (Jupiter up to 115, Saturn up to 293). Always show the date beside the number on the website and re-check before submission.

QUALITATIVE HIGHLIGHTS FOR STORYTELLING

- **Mercury** - smallest planet, extreme temperature swings, cratered like the Moon.
- **Venus** - hottest planet (runaway greenhouse effect), rotates backward.
- **Earth** - only known planet with life, liquid water and a protective atmosphere.
- **Mars** - the Red Planet; iron-oxide dust; evidence of ancient water; target for human exploration.
- **Jupiter** - largest planet; Great Red Spot; strongest magnetic field.
- **Saturn** - iconic rings; least dense planet (would float in water).
- **Uranus** - rotates on its side (~98 degree tilt); pale blue-green from methane.
- **Neptune** - strongest winds; deep blue; found through mathematical prediction.

PLANET 1 OF 8

Mercury

The smallest, swiftest and most extreme planet



PLANET AT A GLANCE

DISTANCE FROM SUN **57.9 million km (0.39 AU)**	DIAMETER **4,879 km (0.38 x Earth)**	DAY (ROTATION) **59 Earth days**	YEAR (ORBIT) **88 Earth days**
MOONS **0**	TEMPERATURE **-180°C to 430°C**	SURFACE GRAVITY **3.7 m/s^2 (0.38 x Earth)**	TYPE **Rocky terrestrial**

THE STORY IN BRIEF

Mercury is the Sun's closest neighbour and the smallest planet - a scorched, crater-covered world only about 40% wider than Earth's Moon. It races around the Sun in just 88 days, yet spins so slowly that one sunrise-to-sunrise day lasts about 176 Earth days. With almost no atmosphere to hold heat, its sunlit side bakes while its night side plunges far below freezing.

QUICK ANSWERS

What is it made of?	Rocky terrestrial planet with a huge metallic core (about 70% metal and 30% silicate rock by mass).
Distance and year	57.9 million km from the Sun; one orbit (a year) takes 88 Earth days. It is the fastest planet, moving at about 47 km per second.
Day length	One rotation takes 59 Earth days; because of its fast orbit, sunrise to sunrise takes about 176 Earth days.
Size vs Earth	4,879 km wide - about 38% of Earth's diameter. The smallest planet in the Solar System.
Moons	None.
Atmosphere and temperature	Virtually no atmosphere (only a thin exosphere). Temperatures swing from about -180°C at night to 430°C by day.
What makes it unique?	Smallest and fastest-orbiting planet, heavily cratered, with the most extreme temperature swings. Water ice hides in shadowed polar craters.
Discovery / first visit	Known since ancient times. First visited by NASA's Mariner 10 (1974-75).
Missions and what we learned	Mariner 10 and NASA's MESSENGER orbiter (2011-2015, mapped the whole planet). ESA/JAXA's BepiColombo is due to enter orbit in late 2026.

COMPARED WITH EARTH

Measure	Mercury	Earth (for reference)
Diameter	4,879 km (0.38 x Earth)	12,742 km
Mass	0.055 x Earth (about 1/18)	1 Earth mass (5.97×10^{24} kg)
Surface gravity	0.38 x Earth	9.8 m/s^2 (1 g)
A 50 kg person would weigh	about 19 kg	50 kg
Distance from Sun	0.39 AU	1 AU (149.6 million km)
Orbital speed	47.4 km/s	29.8 km/s

Did you know? Mercury is shrinking! As its big iron core cools, the whole planet contracts, wrinkling the crust into cliffs that run for hundreds of kilometres.

PLANET 1 OF 8

PAGE 2 OF 2 | LEARN MORE

Mercury

The smallest, swiftest and most extreme planet



INSIDE AND OUT

Mercury's iron-rich core is enormous - roughly 80-85% of the planet's radius - wrapped in a thin rocky mantle and crust. The surface looks a lot like the Moon: ancient craters, smooth volcanic plains and long cliffs. A thin exosphere of atoms blasted off the surface by the solar wind stretches into a faint, comet-like tail.

- **Caloris Basin:** an impact crater about 1,550 km wide, one of the largest in the Solar System.
- **Polar ice:** radar and MESSENGER data show water ice in permanently shadowed polar craters - on the planet nearest the Sun.
- **Weak magnetic field:** only about 1% as strong as Earth's, yet surprising for such a small world.
- **Lobate scarps:** cliffs formed as the planet cooled and shrank.

MOONS – WHY DOES MERCURY HAVE NO MOONS?

No moons. Mercury is small and sits deep in the Sun's gravitational pull. Any moon would be tugged away or pushed into an unstable orbit, so nothing has been able to stay in orbit around it.

Rocky planet	Moons	Why
Mercury	0	Too small and too close to the Sun's strong gravity
Venus	0	Cause debated: solar tides and an unusual spin history
Earth	1	The Moon - likely formed from a giant impact
Mars	2	Phobos and Deimos - small, possibly captured asteroids

EXPLORATION TIMELINE

When	Mission	What it did / found
1974-75	Mariner 10 (NASA)	First spacecraft to reach Mercury; three flybys imaged about 45% of the surface and detected a magnetic field.
2011-15	MESSENGER (NASA)	First to orbit Mercury; mapped the entire planet, confirmed polar ice and revealed that the planet has shrunk.
2018-	BepiColombo (ESA/JAXA)	Two orbiters launched in October 2018; arrival in Mercury orbit is planned for late 2026.

FUN FACTS

- A year on Mercury (88 days) is shorter than one of its days (about 176 Earth days).
- It is closest to the Sun but not the hottest planet - that title belongs to Venus.
- Named after the Roman messenger god because it moves fastest across the sky.

Website idea: Day/night slider that swings the temperature from -180°C to 430°C as the Sun rises.

Scale check: Earth's volume is about 18 times Mercury's; Mercury is only ~40% wider than our Moon.

PLANET 2 OF 8

Venus

Earth's scorching twin, shrouded in acid clouds



PLANET AT A GLANCE

DISTANCE FROM SUN **108.2 million km (0.72 AU)**	DIAMETER **12,104 km (0.95 x Earth)**	DAY (ROTATION) **243 Earth days (backward)**	YEAR (ORBIT) **225 Earth days**
MOONS **0**	TEMPERATURE **about 465°C (surface average)**	SURFACE GRAVITY **8.9 m/s^2 (0.90 x Earth)**	TYPE **Rocky terrestrial**

THE STORY IN BRIEF

Venus is Earth's twin in size but not in climate. Its thick carbon dioxide atmosphere traps heat in a runaway greenhouse effect, making it the hottest planet - hot enough to melt lead - even though Mercury is closer to the Sun. It also spins backward, so the Sun rises in the west, and one day there lasts longer than its year.

QUICK ANSWERS

What is it made of?	Rocky terrestrial planet wrapped in a dense carbon dioxide atmosphere (about 96%) with clouds of sulfuric acid.
Distance and year	108.2 million km from the Sun; one orbit takes 225 Earth days.
Day length	One rotation takes 243 Earth days, and it spins backward (retrograde) compared with most planets - longer than its year.
Size vs Earth	12,104 km wide - about 95% of Earth's diameter, making it Earth's near-twin in size.
Moons	None.
Atmosphere and temperature	Thick CO_2 atmosphere causing a runaway greenhouse effect; the surface averages about 465°C and the pressure is roughly 92 times Earth's.
What makes it unique?	The hottest planet despite not being closest to the Sun; it rotates opposite to its orbit and is the brightest planet in our sky.
Discovery / first visit	Known since antiquity as the 'morning and evening star'. First flown past by Mariner 2 (1962).
Missions and what we learned	Soviet Venera probes (first landing, 1970), NASA's Magellan radar mapper, ESA's Venus Express and Japan's Akatsuki.

COMPARED WITH EARTH

Measure	Venus	Earth (for reference)
Diameter	12,104 km (0.95 x Earth)	12,742 km
Mass	0.82 x Earth	1 Earth mass (5.97 x 10^{24} kg)
Surface gravity	0.90 x Earth	9.8 m/s^2 (1 g)
A 50 kg person would weigh	about 45 kg	50 kg
Distance from Sun	0.72 AU	1 AU (149.6 million km)
Orbital speed	35.0 km/s	29.8 km/s

Did you know? The longest-lived Soviet lander, Venera 13 (1982), survived only about two hours on the surface before the heat and pressure ended its mission.

PLANET 2 OF 8

Venus

Earth's scorching twin, shrouded in acid clouds

PAGE 2 OF 2 | LEARN MORE



INSIDE AND OUT

Beneath its clouds Venus has a rocky mantle and iron core much like Earth's. The surface is mostly volcanic plains with raised highland 'continents'. Radar from NASA's Magellan (1990-94) mapped about 98% of it through the clouds, finding thousands of volcanoes and surprisingly few craters - a sign the surface is geologically young. High-altitude winds circle the whole planet in about four Earth days, far faster than the planet itself rotates.

- **Maxwell Montes:** the tallest mountain range, rising about 11 km above the surrounding plains.
- **Crushing pressure:** surface pressure is about 92 times Earth's - like being roughly 900 m deep in our oceans.
- **Possible active volcanoes:** comparing Magellan radar images has revealed changes consistent with recent volcanic activity.
- **Phases:** like the Moon, Venus shows phases; Galileo's 1610 observations of them supported the Sun-centred model.

MOONS – WHY DOES VENUS HAVE NO MOONS?

No moons. The exact reason is still debated. Scientists think the Sun's strong gravity and tides, combined with Venus's history (possibly an early collision that changed its spin), left no stable orbit for a moon to survive.

Rocky planet	Moons	Why
Mercury	0	Too small and too close to the Sun's strong gravity
Venus	0	Cause debated: solar tides and an unusual spin history
Earth	1	The Moon - likely formed from a giant impact
Mars	2	Phobos and Deimos - small, possibly captured asteroids

EXPLORATION TIMELINE

When	Mission	What it did / found
1962	Mariner 2 (NASA)	First successful flyby of any other planet; measured its extremely hot surface.
1970-82	Venera 7, 9, 13 (USSR)	Venera 7 was the first to transmit from another planet's surface; Venera 9 sent the first surface photo (1975).
1990-94	Magellan (NASA)	Radar-mapped about 98% of the surface, revealing volcanoes, lava plains and few craters.
2015-24	Akatsuki (JAXA)	Studied Venus's clouds, winds and weather from orbit until contact was lost in 2024.
2030s	EnVision (ESA)	Planned orbiter to study Venus's geology and atmosphere; not yet launched.

FUN FACTS

- Venus is the brightest planet in our night sky - only the Sun and Moon outshine it.
- It is the only major planet named after a female deity (the Roman goddess of love).
- Because it spins backward, the Sun rises in the west and sets in the east.

Website idea: A 'descend through the clouds' scroll that shows temperature and pressure climbing toward the surface.

Scale check: Venus is almost Earth's twin: only about 640 km narrower than Earth.

PLANET 3 OF 8

Earth

Our home - the only known world with life



PLANET AT A GLANCE

DISTANCE FROM SUN **149.6 million km (1 AU)**	DIAMETER **12,742 km (reference size)**	DAY (ROTATION) **24 hours**	YEAR (ORBIT) **365.25 days**
MOONS **1 (the Moon)**	TEMPERATURE **about 15°C (average)**	SURFACE GRAVITY **9.8 m/s^2 (1 g)**	TYPE **Rocky terrestrial**

THE STORY IN BRIEF

Earth is the only place we know with life. It orbits in the 'Goldilocks zone' - not too hot, not too cold - where liquid water can exist on the surface. A nitrogen-oxygen atmosphere, a protective magnetic field and a large Moon that helps steady its tilt all make it a remarkably stable home.

QUICK ANSWERS

What is it made of?	Rocky terrestrial planet with a layered structure: an iron-nickel core, a mantle and a thin crust.
Distance and year	149.6 million km from the Sun (1 astronomical unit); one orbit takes 365.25 days - the reason for a leap day every four years.
Day length	One rotation takes 24 hours (23 hours 56 minutes measured against the stars).
Size vs Earth	12,742 km diameter - the reference point used to compare every other planet; the largest of the four rocky planets.
Moons	One: the Moon.
Atmosphere and temperature	About 78% nitrogen and 21% oxygen; the average surface temperature is about 15°C, and liquid water covers roughly 71% of the surface.
What makes it unique?	The only known planet to support life, protected by a magnetic field that shields it from harmful solar radiation; it also has active plate tectonics.
Discovery / first visit	Our home planet - studied continuously throughout human history. Sputnik 1 (1957) was the first satellite to orbit it.
Missions and what we learned	Explored from space by satellites, the International Space Station and countless Earth-observation missions.

COMPARED WITH THE MOON

Measure	Earth	The Moon (for reference)
Diameter	12,742 km	3,474 km (0.27 x Earth)
Mass	1 Earth mass (5.97×10^{24} kg)	0.012 x Earth
Surface gravity	9.8 m/s^2 (1 g)	1.6 m/s^2 (0.17 g)
A 50 kg person would weigh	50 kg	about 8 kg
Distance	1 AU from the Sun (149.6 million km)	384,400 km from Earth
Orbital speed	29.8 km/s around the Sun	1.0 km/s around Earth

Did you know? Earth is not a perfect sphere: its spin makes it bulge, so it is about 42 km wider across the equator than from pole to pole.

PLANET 3 OF 8

Earth

Our home - the only known world with life

PAGE 2 OF 2 | LEARN MORE



INSIDE AND OUT

Earth's interior has four main layers: a solid inner core of iron and nickel, a liquid outer core that generates the magnetic field, a thick mantle of hot rock, and a thin crust only 5-70 km deep. The crust is split into moving plates that build mountains, trigger earthquakes and recycle the surface. Above it, a thin atmosphere protects and warms the planet.

- **Magnetosphere:** deflects the solar wind and produces auroras near the poles.
- **Plate tectonics:** Earth is the only planet known with this active, surface-recycling process.
- **Water world:** about 71% of the surface is ocean, and most of Earth's water is salty.
- **Ozone layer:** absorbs most of the Sun's harmful ultraviolet radiation.

MOONS - OUR MOON AT A GLANCE

The Moon	Value	Details
Size	3,474 km	About 27% of Earth's diameter - large compared with its planet.
Distance	384,400 km	Average distance; the Moon drifts away from Earth by about 3.8 cm each year.
Orbit	27.3 days	Orbit around Earth; the cycle of phases takes 29.5 days.
Origin	~4.5 billion years	Leading theory: a giant impact by a Mars-sized body (Theia) early in Earth's history.
Effects	Tides, steady tilt	Its gravity drives ocean tides and helps keep Earth's axial tilt stable. It is tidally locked: we always see the same face.
Visits	Apollo 1969-72	Twelve astronauts have walked on it. India's Chandrayaan-3 landed near the lunar south pole in August 2023.

EXPLORATION TIMELINE

When	Mission	What it did / found
1957	Sputnik 1 (USSR)	The first artificial satellite, starting the Space Age.
1960	TIROS-1 (NASA)	First weather satellite, showing Earth's cloud patterns from space.
1975	Aryabhata (ISRO)	India's first satellite; ISRO's later missions monitor weather, oceans and land.
1972-	Landsat (NASA/USGS)	The longest continuous record of Earth's land from space.
1998-	International Space Station	Continuously crewed since 2000; research in microgravity and Earth observation.

FUN FACTS

- Earth is the densest planet in the Solar System.
- All of Earth's water gathered into one ball would be only about 1,400 km wide.
- It is the only planet not named after a Greek or Roman god - its name comes from old words for 'ground'.

Website idea: A layered cut-away that peels from atmosphere to core, with a day/night line sweeping across the globe.

Scale check: About 1,300 Earths would fit inside Jupiter, and about 109 Earths would line up across the Sun.

PLANET 4 OF 8

Mars

The rusty Red Planet with a watery past



PLANET AT A GLANCE

DISTANCE FROM SUN **227.9 million km (1.52 AU)**	DIAMETER **6,779 km (0.53 x Earth)**	DAY (ROTATION) **24.6 hours (a 'sol')**	YEAR (ORBIT) **687 Earth days**
MOONS **2**	TEMPERATURE **about -63°C (average)**	SURFACE GRAVITY **3.7 m/s^2 (0.38 x Earth)**	TYPE **Rocky terrestrial**

THE STORY IN BRIEF

Mars is a cold, dusty desert world painted red by iron oxide - rust. Yet its canyons, dry riverbeds and delta deposits show that water once flowed across its surface. Today it is a favourite target for robotic explorers and the leading destination for future human missions beyond the Moon.

QUICK ANSWERS

What is it made of?	Rocky terrestrial planet with an iron-oxide-rich surface, giving it a reddish colour.
Distance and year	227.9 million km from the Sun; one orbit takes 687 Earth days.
Day length	One rotation takes about 24.6 hours - a Martian day (a 'sol') is very close to an Earth day.
Size vs Earth	6,779 km wide - about 53% of Earth's diameter.
Moons	Two small moons: Phobos and Deimos.
Atmosphere and temperature	Thin atmosphere, mostly carbon dioxide (about 95%), with under 1% of Earth's surface pressure; the average temperature is around -63°C.
What makes it unique?	The 'Red Planet': strong evidence of ancient rivers and lakes, Olympus Mons (the largest volcano in the Solar System) and the Valles Marineris canyon system.
Discovery / first visit	Known since antiquity. First successfully flown past by Mariner 4 (1965).
Missions and what we learned	Viking landers, many orbiters and rovers including Curiosity and Perseverance; India's Mangalyaan reached orbit in 2014.

COMPARED WITH EARTH

Measure	Mars	Earth (for reference)
Diameter	6,779 km (0.53 x Earth)	12,742 km
Mass	0.11 x Earth	1 Earth mass (5.97×10^{24} kg)
Surface gravity	0.38 x Earth	9.8 m/s^2 (1 g)
A 50 kg person would weigh	about 19 kg	50 kg
Distance from Sun	1.52 AU	1 AU (149.6 million km)
Orbital speed	24.1 km/s	29.8 km/s

Did you know? Sunsets on Mars look blue. Fine dust in the thin atmosphere scatters light differently than on Earth, so the sky around the setting Sun glows blue.

PLANET 4 OF 8

PAGE 2 OF 2 | LEARN MORE

Mars

The rusty Red Planet with a watery past



INSIDE AND OUT

Mars has a metal-rich core, a rocky mantle and a crust. Unlike Earth it lost its global magnetic field billions of years ago, and the solar wind slowly stripped away much of its atmosphere. Its surface holds volcanic regions, enormous canyons, polar caps of water ice and frozen carbon dioxide, and planet-wide dust storms that can last for weeks. Minerals such as clays and river-delta deposits at Jezero Crater show that liquid water was once stable there.

- **Olympus Mons:** a shield volcano about 22 km high - nearly 2.5 times the height of Mount Everest.
- **Valles Marineris:** a canyon roughly 4,000 km long and up to 7 km deep - about the width of the United States.
- **Polar caps:** layers of water ice topped with frozen carbon dioxide that grow and shrink with the seasons.
- **Possible ancient life:** Perseverance is collecting rock samples, including one with features that could be a potential biosignature - still under study.

MOONS – MARS'S TWO SMALL MOONS

Moon	Size	Details
Phobos	22 km average (27 x 22 x 18 km)	Larger and closer: it orbits only about 6,000 km above the surface, circling Mars three times a day (7 h 39 min). It is slowly spiralling inward and may break apart in about 50 million years. Its big Stickney crater is about 9 km wide.
Deimos	12 km average (15 x 12 x 10 km)	Smaller and farther out, taking about 30 hours per orbit. Its surface is smoother because dust fills many craters.
Origin	Debated	Both may be captured asteroids, or rebuilt from debris thrown up by a giant impact. They were discovered in 1877 by Asaph Hall and named after Fear and Panic, companions of the war god Ares.

EXPLORATION TIMELINE

When	Mission	What it did / found
1965	Mariner 4 (NASA)	First successful flyby; returned 22 photos showing a cratered, Moon-like surface.
1976	Viking 1 and 2 (NASA)	First long-lived landers; searched the soil for signs of life.
2004-18	Spirit and Opportunity (NASA)	Rovers that found minerals formed in water; Opportunity worked for almost 15 years.
2012-	Curiosity (NASA)	Found that Gale Crater once had conditions suitable for microbial life.
2014	Mangalyaan / MOM (ISRO)	India became the first nation to reach Mars orbit on its first attempt.
2021-	Perseverance and Ingenuity (NASA)	Jezero Crater rover caching samples; Ingenuity made the first powered flights on another world.

FUN FACTS

- A Mars year lasts 687 Earth days - about 668 sols.
- Phobos and Deimos are named after the Greek words for fear and panic.
- Mars has only about 15% of Earth's volume and about 11% of its mass.

Website idea: A 'then and now' toggle switching between today's dry red planet and an artist's view of ancient lakes.

Scale check: Mars is roughly half Earth's width, and its volcano Olympus Mons is about 600 km across.

PLANET 5 OF 8

Jupiter

King of the planets - a miniature solar system



PLANET AT A GLANCE

DISTANCE FROM SUN **778.5 million km (5.2 AU)**	DIAMETER **139,820 km (about 11 x Earth)**	DAY (ROTATION) **9.9 hours**	YEAR (ORBIT) **11.9 Earth years**
MOONS **100+ (101 confirmed, 2026)**	TEMPERATURE **about -110°C (cloud tops)**	SURFACE GRAVITY **24.8 m/s^2 (2.5 x Earth)**	TYPE **Gas giant**

THE STORY IN BRIEF

Jupiter is the king of the planets: more than twice as massive as all the others combined, and so large that about 1,300 Earths could fit inside. It has no solid surface - only swirling bands of hydrogen and helium clouds, studded with storms like the Great Red Spot. With more than a hundred moons, it is almost a miniature solar system of its own.

QUICK ANSWERS

What is it made of?	Gas giant made mainly of hydrogen and helium, with no solid surface.
Distance and year	778.5 million km from the Sun; one orbit takes 11.9 Earth years.
Day length	Just 9.9 hours - the fastest rotation of any planet, which flattens it slightly at the poles.
Size vs Earth	139,820 km across - the largest planet, about 11 times Earth's diameter.
Moons	100+ known (101 confirmed in 2026), including the four large Galilean moons: Io, Europa, Ganymede and Callisto.
Atmosphere and temperature	Hydrogen-helium atmosphere arranged in coloured bands, with the Great Red Spot storm; cloud tops are about -110°C.
What makes it unique?	The largest planet, with the strongest magnetic field of any planet and a faint ring system.
Discovery / first visit	Known since antiquity. Galileo discovered its four largest moons in 1610.
Missions and what we learned	Pioneer 10/11, Voyager 1 and 2, the Galileo orbiter and Juno (in orbit since 2016); Europa Clipper and JUICE are on their way.

COMPARED WITH EARTH

Measure	Jupiter	Earth (for reference)
Diameter	139,820 km (about 11 x Earth)	12,742 km
Mass	318 x Earth	1 Earth mass (5.97 x 10^{24} kg)
Gravity at cloud tops	2.5 x Earth	9.8 m/s^2 (1 g)
A 50 kg person would weigh	about 126 kg (at cloud tops)	50 kg
Distance from Sun	5.2 AU	1 AU (149.6 million km)
Orbital speed	13.1 km/s	29.8 km/s

Did you know? The Great Red Spot is bigger than Earth - about 1.3 Earths wide today - and astronomers have watched it for at least 150 years, though it is slowly shrinking.

PLANET 5 OF 8

PAGE 2 OF 2 | LEARN MORE

Jupiter

King of the planets - a miniature solar system



INSIDE AND OUT

Jupiter's hydrogen and helium gas grows denser with depth, turning into a liquid and then into electrically conducting 'metallic hydrogen' that drives its enormous magnetic field. Whether a solid core sits at the centre is still debated. The cloud bands are light 'zones' of rising gas and dark 'belts' of sinking gas, separated by jet streams that blow in opposite directions. Jupiter also gives off more heat than it receives from the Sun.

- **Great Red Spot:** a giant storm, wider than Earth, watched for at least 150 years.
- **Magnetosphere:** about 20,000 times stronger than Earth's and extending millions of kilometres into space.
- **Rings:** faint and dusty, discovered by Voyager 1 in 1979.
- **Polar cyclones:** Juno revealed clusters of huge storms circling each pole.

MOONS - THE FOUR GALILEAN MOONS (DISCOVERED 1610)

Moon	Diameter	Details
Io	3,643 km	The most volcanically active body in the Solar System, with 400+ active volcanoes. Orbit: 1.8 days.
Europa	3,122 km	An ice shell hides a salty global ocean - a top target in the search for life. Orbit: 3.6 days.
Ganymede	5,268 km	The largest moon in the Solar System - bigger than Mercury - and the only moon with its own magnetic field. Orbit: 7.2 days.
Callisto	4,821 km	The most heavily cratered surface known; very old, and possibly has a buried ocean. Orbit: 16.7 days.

EXPLORATION TIMELINE

When	Mission	What it did / found
1973-74	Pioneer 10 and 11 (NASA)	First spacecraft to fly past Jupiter; measured its radiation belts.
1979	Voyager 1 and 2 (NASA)	Discovered Io's volcanoes and Jupiter's faint rings; detailed images of the Great Red Spot.
1995-2003	Galileo (NASA)	First Jupiter orbiter; dropped a probe into the atmosphere and found evidence of a subsurface ocean on Europa.
2016-	Juno (NASA)	Polar orbiter mapping gravity, magnetism and the deep atmosphere; revealed polar cyclones.
2024-31	Europa Clipper, JUICE	NASA's Europa Clipper (launched 2024) and ESA's JUICE (launched 2023) are heading to the icy moons.

FUN FACTS

- Jupiter's mass is about 318 Earths - about 2.5 times all other planets combined.
- A day lasts under 10 hours, but a year lasts almost 12 Earth years.
- Io, Europa and Ganymede are locked in a 1:2:4 orbital dance called a resonance.

Website idea: A spin control showing the 10-hour day, with the four Galilean moons circling in real time.

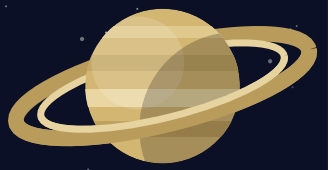
Scale check: If Jupiter were a basketball, Earth would be about the size of a grape (roughly 2 cm).

PLANET 6 OF 8

PAGE 1 OF 2 | SUMMARY

Saturn

The ringed jewel that would float in water



PLANET AT A GLANCE

DISTANCE FROM SUN	DIAMETER	DAY (ROTATION)	YEAR (ORBIT)
1,434 million km (9.6 AU)	**116,460 km (about 9 x Earth)**	**10.7 hours**	**29.5 Earth years**
MOONS	**TEMPERATURE**	**SURFACE GRAVITY**	**TYPE**
285+ (confirmed, 2026)	**about -178°C (cloud tops)**	**10.4 m/s^2 (1.06 x Earth)**	**Gas giant**

THE STORY IN BRIEF

Saturn is the jewel of the Solar System, famous for a brilliant ring system made of billions of pieces of ice and rock. It is so light for its size that it would float in a bathtub big enough to hold it. With hundreds of moons - including Titan, with a thick atmosphere, and Enceladus, with icy geysers - it is one of the best places to search for the ingredients of life beyond Earth.

QUICK ANSWERS

What is it made of?	Gas giant made mainly of hydrogen and helium, with no solid surface; its average density is lower than water's.
Distance and year	1,434 million km from the Sun; one orbit takes 29.5 Earth years.
Day length	One rotation takes about 10.7 hours.
Size vs Earth	116,460 km across - the second-largest planet, about 9 times Earth's diameter.
Moons	285+ confirmed (March 2026) - more than any other planet - notably Titan (thick atmosphere, methane lakes) and Enceladus (icy geysers).
Atmosphere and temperature	Hydrogen-helium atmosphere with pale bands and a six-sided jet stream at the north pole; cloud tops are about -178°C.
What makes it unique?	Its iconic rings make it instantly recognisable, and it is the least dense planet - it would float in water.
Discovery / first visit	Known since antiquity. Galileo saw odd 'handles' in 1610; Christiaan Huygens correctly identified the rings in 1655.
Missions and what we learned	Pioneer 11, Voyager 1 and 2, and extensively by Cassini-Huygens (2004-2017).

COMPARED WITH EARTH

Measure	Saturn	Earth (for reference)
Diameter	116,460 km (about 9 x Earth)	12,742 km
Mass	95 x Earth	1 Earth mass (5.97 x 10^{24} kg)
Gravity at cloud tops	1.06 x Earth	9.8 m/s^2 (1 g)
A 50 kg person would weigh	about 53 kg (at cloud tops)	50 kg
Distance from Sun	9.6 AU	1 AU (149.6 million km)
Orbital speed	9.7 km/s	29.8 km/s

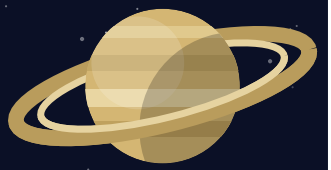
Did you know? Saturn's rings are about 280,000 km wide yet often only about 10 metres thick in places - proportionally thinner than a sheet of paper.

PLANET 6 OF 8

PAGE 2 OF 2 | LEARN MORE

Saturn

The ringed jewel that would float in water



INSIDE AND OUT

Saturn's gas thickens with depth into liquid and then metallic hydrogen, around a core of heavier elements. The rings, labelled A to G, are more than 99% water ice - particles from dust grains to house-sized boulders. Their age is debated. Because Saturn is tilted about 27 degrees, we see the rings from different angles over its 29-year orbit; they appeared edge-on and nearly vanished from Earth in 2025. Giant 'Great White Spot' storms erupt roughly every 30 years.

- **Rings:** seven main ring groups; the A and B rings are separated by the Cassini Division, a gap about 4,800 km wide.
- **Hexagon storm:** a six-sided jet stream at the north pole, wider than two Earths.
- **Low density:** about 0.69 g/cm^3, less than water's 1.0.
- **Moon explosion:** 128 new moons were announced in 2025 alone.

MOONS - SATURN'S 285+ MOONS - FIVE TO KNOW (MOST OTHERS ARE TINY, IRREGULAR ROCKS)

Moon	Diameter	Details
Titan	5,150 km	Second-largest moon in the Solar System, bigger than Mercury. Thick nitrogen atmosphere and lakes of liquid methane and ethane. The Huygens probe landed on it in 2005; NASA's Dragonfly rotorcraft is planned.
Enceladus	504 km	Geysers of water ice erupt from cracks near its south pole, feeding Saturn's E ring. A salty ocean lies below - a prime target for astrobiology.
Mimas	396 km	The giant Herschel crater makes it look like the Death Star.
Rhea	1,527 km	Saturn's second-largest moon; mostly water ice and heavily cratered.
Iapetus	1,469 km	Two-toned: one hemisphere dark, the other bright, with a mountain ridge along its equator.

EXPLORATION TIMELINE

When	Mission	What it did / found
1979	Pioneer 11 (NASA)	First flyby; discovered the thin F ring and a new moon.
1980-81	Voyager 1 and 2 (NASA)	Detailed images of the rings, Titan's haze and the north-pole hexagon.
2004-17	Cassini-Huygens (NASA/ESA/ASI)	Orbited for 13 years; Huygens landed on Titan; found Enceladus's plumes; ended with a planned plunge in 2017.
2028-34	Dragonfly (NASA)	Planned rotorcraft to fly across Titan's surface; launch planned 2028, arrival 2034.

FUN FACTS

- Saturn has more moons than all other planets' moons combined (Jupiter 101 + Uranus 29 + Neptune 16 + Mars 2 + Earth 1 = 149).
- Its rapid spin flattens it: Saturn is the most oblate planet.
- Named after the Roman god of agriculture; Saturday is named for it.

Website idea: Drag to tilt the planet and watch the rings open up and slide edge-on, with moon markers that pop up.

Scale check: Saturn's main rings span about three-quarters of the distance from Earth to the Moon.

PLANET 7 OF 8

Uranus

The ice giant that rolls around the Sun on its side



PLANET AT A GLANCE

DISTANCE FROM SUN **2,871 million km (19.2 AU)**	DIAMETER **50,724 km (about 4 x Earth)**	DAY (ROTATION) **17.2 hours (backward)**	YEAR (ORBIT) **84 Earth years**
MOONS **29 (known, 2025)**	TEMPERATURE **about -195°C (as low as -224°C)**	SURFACE GRAVITY **8.9 m/s^2 (0.90 x Earth)**	TYPE **Ice giant**

THE STORY IN BRIEF

Uranus is the ice giant that rolls around the Sun on its side. Its axis is tilted about 98 degrees, so each pole gets 42 years of continuous sunlight followed by 42 years of darkness. Methane in its atmosphere absorbs red light and gives it a pale blue-green colour. It has faint rings and 29 known moons named after characters from Shakespeare and Alexander Pope.

QUICK ANSWERS

What is it made of?	Ice giant: a mix of 'icy' materials (water, ammonia, methane) around a small rocky core, wrapped in hydrogen and helium.
Distance and year	2,871 million km from the Sun; one orbit takes 84 Earth years.
Day length	One rotation takes about 17.2 hours, and it spins on its side.
Size vs Earth	50,724 km across - about 4 times Earth's diameter.
Moons	29 known, including the five large moons Miranda, Ariel, Umbriel, Titania and Oberon.
Atmosphere and temperature	Hydrogen, helium and methane atmosphere that gives it a pale blue-green colour; extremely cold, the coldest planetary atmosphere measured.
What makes it unique?	Rotates almost on its side, with an axial tilt of about 98 degrees - likely the result of an ancient massive collision.
Discovery / first visit	Discovered by William Herschel in 1781 - the first planet found with a telescope.
Missions and what we learned	Visited only once, by Voyager 2 during its 1986 flyby; no dedicated mission has been sent since.

COMPARED WITH EARTH

Measure	Uranus	Earth (for reference)
Diameter	50,724 km (about 4 x Earth)	12,742 km
Mass	14.5 x Earth	1 Earth mass (5.97 x 10^{24} kg)
Gravity at cloud tops	0.90 x Earth	9.8 m/s^2 (1 g)
A 50 kg person would weigh	about 45 kg (at cloud tops)	50 kg
Distance from Sun	19.2 AU	1 AU (149.6 million km)
Orbital speed	6.8 km/s	29.8 km/s

Did you know? When William Herschel first spotted Uranus he thought it was a comet. He even proposed naming it after King George III!

PLANET 7 OF 8

PAGE 2 OF 2 | LEARN MORE

Uranus

The ice giant that rolls around the Sun on its side



INSIDE AND OUT

Uranus has a small rocky core, a huge mantle of hot, dense 'icy' fluid - water, ammonia and methane under enormous pressure - and an outer layer of hydrogen and helium. 'Ice' here means materials that freeze easily, not ice cubes. Each of its four seasons lasts about 21 years. Its magnetic field is lopsided, tilted about 59 degrees from the spin axis and offset from the planet's centre. Unlike Neptune, Uranus gives off very little internal heat.

- **Extreme tilt:** about 98 degrees, probably from an ancient giant collision.
- **Rings:** 13 known, mostly narrow and dark; first detected in 1977 when Uranus passed in front of a star.
- **Crooked magnetic field:** tilted and offset, so it wobbles as the planet spins.
- **Blue-green haze:** methane gas absorbs red light, leaving a pale cyan colour.

MOONS – FIVE LARGE MOONS (NAMED AFTER SHAKESPEARE AND POPE CHARACTERS)

Moon	Diameter	Details
Miranda	472 km	A patchwork surface of cliffs and canyons. Verona Rupes may be a cliff up to about 20 km high - possibly the tallest in the Solar System.
Ariel	1,158 km	The brightest and youngest-looking big moon, with deep canyons and smooth plains.
Umbriel	1,169 km	The darkest of the large moons; ancient and cratered.
Titania	1,578 km	The largest Uranian moon, with huge canyon systems; discovered by Herschel in 1787.
Oberon	1,523 km	Old and heavily cratered, with dark material on some crater floors.

EXPLORATION TIMELINE

When	Mission	What it did / found
1986	Voyager 2 (NASA)	Only flyby (24 Jan 1986); found 10 new moons and two new rings and measured the magnetic field.
2022-25	James Webb Space Telescope	Sharp infrared images of its rings; discovered a new small moon in 2025 (the 29th).
2030s?	Uranus Orbiter and Probe	Top-priority flagship recommended by US planetary scientists; not yet launched.

FUN FACTS

- Its 84-year orbit means most people see less than one Uranian year in a lifetime.
- It was the first planet discovered with a telescope rather than seen by ancient observers.
- Named after the Greek sky god - the only planet named from Greek rather than Roman mythology.

Website idea: A tilt toggle that shows how sunlight strikes Uranus's poles over its 84-year orbit.

Scale check: About 63 Earths could fit inside Uranus by volume.

PLANET 8 OF 8

Neptune

The distant blue world of supersonic winds



PLANET AT A GLANCE

DISTANCE FROM SUN **4,495 million km (30.1 AU)**	DIAMETER **49,244 km (about 3.9 x Earth)**	DAY (ROTATION) **16.1 hours**	YEAR (ORBIT) **165 Earth years**
MOONS **16 (known, 2026)**	TEMPERATURE **about -201°C (cloud tops)**	SURFACE GRAVITY **11.2 m/s^2 (1.14 x Earth)**	TYPE **Ice giant**

THE STORY IN BRIEF

Neptune is the farthest planet, a deep-blue ice giant so distant that sunlight takes over four hours to reach it. It has the strongest winds in the Solar System, reaching about 2,100 km/h, and its giant moon Triton orbits backward. It was the first planet found by mathematics before anyone saw it through a telescope.

QUICK ANSWERS

What is it made of?	Ice giant with a composition similar to Uranus: water, ammonia and methane ices around a rocky core, plus hydrogen and helium.
Distance and year	4,495 million km from the Sun; one orbit takes 165 Earth years.
Day length	One rotation takes about 16.1 hours.
Size vs Earth	49,244 km across - slightly smaller than Uranus but more massive (about 17 Earth masses).
Moons	16 known, notably Triton, which orbits in the opposite direction to Neptune's rotation.
Atmosphere and temperature	Hydrogen, helium and methane atmosphere; the strongest winds in the Solar System, about 2,100 km/h; around -201°C.
What makes it unique?	Known for its deep blue colour and for being the only planet discovered through mathematical prediction rather than direct observation.
Discovery / first visit	Discovered in 1846 by Johann Galle, using calculations by Urbain Le Verrier that predicted its position from irregularities in Uranus's orbit.
Missions and what we learned	Visited only once, by Voyager 2 during its 1989 flyby; no dedicated mission has been sent since.

COMPARED WITH EARTH

Measure	Neptune	Earth (for reference)
Diameter	49,244 km (about 3.9 x Earth)	12,742 km
Mass	17 x Earth	1 Earth mass (5.97 x 10^{24} kg)
Gravity at cloud tops	1.14 x Earth	9.8 m/s^2 (1 g)
A 50 kg person would weigh	about 57 kg (at cloud tops)	50 kg
Distance from Sun	30.1 AU	1 AU (149.6 million km)
Orbital speed	5.4 km/s	29.8 km/s

Did you know? Neptune completed its first full orbit since its discovery in 2011 - its year is so long that it took 165 years!

PLANET 8 OF 8

PAGE 2 OF 2 | LEARN MORE

Neptune

The distant blue world of supersonic winds



INSIDE AND OUT

Neptune's interior resembles Uranus's: a small rocky core, a deep mantle of hot 'icy' fluid, and an atmosphere of hydrogen, helium and a little methane. Unlike Uranus, it gives off about 2.6 times more energy than it receives from the Sun, and that heat helps power its ferocious winds. Its blue appearance comes partly from methane; a 2024 study showed Neptune's true colour is closer to a greenish blue than the deep blue in many images.

- **Fast winds:** up to about 2,100 km/h - faster than the speed of sound on Earth.
- **Dark spots:** storm systems such as Voyager 2's Great Dark Spot (1989) that appear and vanish.
- **Rings:** five main rings, with clumpy arcs in the outermost Adams ring.
- **Internal heat:** radiates about 2.6 times the energy it receives from the Sun.

MOONS – NEPTUNE'S 16 MOONS – FOUR TO KNOW

Moon	Diameter	Details
Triton	2,707 km	Largest moon, discovered just 17 days after Neptune. It orbits backward - probably a captured Kuiper Belt object. Nitrogen geysers erupt on a surface about -235°C. It is slowly spiralling inward.
Proteus	~420 km	Second-largest moon; dark and irregular, discovered by Voyager 2 in 1989.
Nereid	~340 km	Follows one of the most stretched-out orbits of any moon (about 360 days); discovered in 1949 by Gerard Kuiper.
Hippocamp	~35 km	Tiny moon found by Hubble in 2013; probably debris from a past impact on Proteus.

EXPLORATION TIMELINE

When	Mission	What it did / found
1989	Voyager 2 (NASA)	Only flyby (25 Aug 1989); discovered the Great Dark Spot, six new moons, rings and Triton's geysers.
2022-	James Webb Space Telescope	Crisp infrared images of Neptune's rings and atmosphere.
Future	Proposed concepts	Dedicated orbiter ideas (such as Neptune Odyssey) exist, but no mission has been approved.

FUN FACTS

- Le Verrier's calculations placed Neptune within about one degree of where Galle found it.
- Galileo probably saw Neptune in 1612-13 but recorded it as a faint star.
- Sunlight takes about 4.2 hours to reach Neptune, against about 8 minutes for Earth.

Website idea: A 'wind speed race' comparing the fastest hurricane winds on Earth with Neptune's supersonic gusts.

Scale check: Neptune is about 30 times farther from the Sun than Earth.

04 Interactive and Design Research

How to turn the research into a zoom-scroll experience

How have other space websites handled zoom-scroll storytelling?	Many space-education sites use **scrollytelling** - scrolling triggers animations, zooms or transitions - paired with **parallax** backgrounds to create depth and movement through space, keeping the scale of the universe intuitive.
What tools support parallax or scroll-based zoom?	**Three.js** renders 3D scenes (a rotating galaxy or planets) in the browser, while **GSAP ScrollTrigger** ties animations and camera movement to scroll position, making a smooth zoom-in narrative achievable with web technologies.
What scale comparisons keep it real?	True-to-scale space is mostly empty, so use relatable comparisons (for example, 'if the Sun were a basketball, Earth would be a peppercorn about 30 metres away') or compressed, non-linear scale transitions so the journey never feels boring.

SUGGESTED SITE FLOW (HOW THE STAGES MAP TO THE SCREEN)

Stage	Visual / zoom	Key message	Extra depth
1 Milky Way	Full-screen galaxy view; slow rotation	We live inside a barred spiral galaxy of 100-400 billion stars.	Sagittarius A*, Local Group
2 Solar System	Zoom to the Orion Arm, then the orbits	The Sun holds 99.86% of the mass and keeps everything in orbit.	Nebular hypothesis, scale comparisons
3 The Sun	Bright star fills the screen; solar wind particles	Fusion powers life on Earth.	Sun's life cycle, flares and sunspots
4 Planets	Pan from Mercury outward; each planet opens a card	Eight very different worlds.	Summary page, then Learn More page

INTERACTION IDEAS COLLECTED FROM THE PLANET PAGES

- **Mercury:** Day/night slider that swings the temperature from -180°C to 430°C as the Sun rises.
- **Venus:** A 'descend through the clouds' scroll that shows temperature and pressure climbing toward the surface.
- **Earth:** A layered cut-away that peels from atmosphere to core, with a day/night line sweeping across the globe.
- **Mars:** A 'then and now' toggle switching between today's dry red planet and an artist's view of ancient lakes.
- **Jupiter:** A spin control showing the 10-hour day, with the four Galilean moons circling in real time.
- **Saturn:** Drag to tilt the planet and watch the rings open up and slide edge-on, with moon markers that pop up.
- **Uranus:** A tilt toggle that shows how sunlight strikes Uranus's poles over its 84-year orbit.
- **Neptune:** A 'wind speed race' comparing the fastest hurricane winds on Earth with Neptune's supersonic gusts.

05 Project Name, Sources and Notes

Naming rationale and where the numbers come from

CHOSEN NAME

The Galactic Canvas

This name reflects the site's core concept - presenting the vastness of the galaxy, the Solar System and the planets as an unfolding visual and narrative canvas for the viewer to explore.

SOURCES AND VERIFICATION NOTES

- Stages 1-3 and the original planet answers come from the project's first research compilation.
- Moon counts were refreshed from 2026 sources: NASA Space Place ('How Many Moons Does Each Planet Have?', updated July 2026), EarthSky (March 2026 Minor Planet Center announcement) and Live Science.

- Planetary measurements are rounded values consistent with NASA's planetary fact sheets. Double-check any figure you will quote on screen against NASA Solar System Exploration or ESA pages.
- Fast-changing items to re-check before final submission: moon counts, BepiColombo's arrival at Mercury (planned late 2026), and the status of planned missions such as Dragonfly, EnVision and Europa Clipper.
- Diameters use mean values (Jupiter 139,820 km, Saturn 116,460 km), so ratios to Earth may differ slightly from equatorial figures.